

A 71-POINT ARC REFUTES THE PRINTED FINITE- p FORM OF THE ENTROPIC CAUCHY–DAVENPORT CANDIDATE OF GAVALAKIS, GOH AND KONTOYIANNIS

ABSTRACT. Gavalakis, Goh and Kontoyiannis suggest, in a hedged display whose first line is the only part at issue here, that a candidate entropic Cauchy–Davenport theorem “might be” that

$$H(X + X') \geq \min\{\log p, \log((2^{H(X)+1} - 1)/\sqrt{2})\}$$

for every prime p , every $\mathbb{F}_p$ -valued X and X' an independent copy of X , the logarithm being to base 2. We refute that first line read literally at finite p , in the branch where the minimum is *not* the cap $\log p$: take $p = 101$ and X uniform on the arc $S = \{0, 1, \dots, 70\}$. Then $2^{H(X)+1} - 1 = 141$ exactly, the integer comparison $141^2 = 19881 < 20402 = 2 \cdot 101^2$ shows that the active branch is $\log(141/\sqrt{2})$ and not $\log 101$, and $H(X + X') < \log(141/\sqrt{2})$. The decisive step is a single comparison between two explicit positive integers; no logarithm is evaluated and no floating-point arithmetic is used. Six further witnesses at $p = 61, 151, 211, 307, 401, 509$ are certified the same way. Only the finite- p first line of the display is affected: the asymptotic second line, which carries an $\mathcal{O}_{p \rightarrow \infty}(1)$, is untouched, and nothing here contradicts the authors’ own theorem, an ε -weakened statement restricted to an entropy window: we do not determine that window’s constant, and what we check is that all seven witnesses satisfy the theorem’s conclusion for every $\varepsilon > 0.023$.

1. THE STATEMENT

Throughout, $\log$ is the logarithm to base 2, entropies are in bits, and $0 \log 0 = 0$, as in the source paper [1]. For a prime p we write $\mathbb{F}_p$ for the field with p elements, and X' always denotes an independent copy of the $\mathbb{F}_p$ -valued random variable X .

Gavalakis, Goh and Kontoyiannis [1] discuss what an entropic analogue of the Cauchy–Davenport theorem $|A + B| \geq \min\{p, |A| + |B| - 1\}$ could say. Having observed that the naive analogue fails, they write “In view of the above comments, a more appropriate candidate for an entropic Cauchy–Davenport theorem might be that”, and then display, as equation (2) of

2020 *Mathematics Subject Classification.* 94A17, 11B13, 11B30, 05B10.

Key words and phrases. Shannon entropy, Cauchy–Davenport theorem, sumset, entropy power inequality, prime field, counterexample.

[1],

$$\begin{aligned} H(X + X') &\geq \min \left\{ \log p, \log \left(\frac{2^{H(X)+1} - 1}{\sqrt{2}} \right) \right\} \\ &= \min \left\{ \log p, H(X) + \frac{1}{2} - o_{p \rightarrow \infty}(1) \right\}. \end{aligned}$$

The first line of that display, which is the only part at issue here, reads

$$(1) \quad H(X + X') \geq \min \left\{ \log p, \log \left(\frac{2^{H(X)+1} - 1}{\sqrt{2}} \right) \right\}.$$

Read literally, the quantifiers are the ones the display carries and no others: every prime p , every $\mathbb{F}_p$ -valued X , and X' an independent copy; there is no largeness hypothesis on p , no ε , and no restriction on $H(X)$. That literal finite- p reading is the only thing tested below. We do not claim to have established from the surrounding text what scope the authors intended for the display, and the second line of it carries an $o_{p \rightarrow \infty}(1)$, so an asymptotically intended reading is consistent with everything proved here.

Two points bear on what is being refuted. First, (1) is a hedged candidate display: the authors write “might be”, they do not call it a conjecture, and we do not either. It is nevertheless a definite mathematical assertion, and they refer back to it, describing their first result as a weaker form of it, so it is a statement one may test. Second, what the authors actually *prove*, via a theorem of Tao [2], is the ε -weakened, two-sided-window assertion

$$(2) \quad H(X + X') \geq H(X) + \frac{1}{2} - \varepsilon \quad \text{whenever } k_\varepsilon < H(X) < \log p - K_\varepsilon,$$

a window their prose says “forbids, e.g., deterministic random variables and random variables uniformly distributed on the whole of $\mathbb{F}$ ”. That gap — (1) has no ε and no window — is what the witness below occupies. Nothing in this note touches (2); see Section 5.

Theorem 1. *Let $p = 101$, let $S = \{0, 1, \dots, 70\} \subset \mathbb{F}_{101}$, let X be uniform on S and let X' be an independent copy of X . Then*

$$\min \left\{ \log p, \log \left(\frac{2^{H(X)+1} - 1}{\sqrt{2}} \right) \right\} = \log \left(\frac{141}{\sqrt{2}} \right) < \log 101,$$

and

$$H(X + X') < \log \left(\frac{141}{\sqrt{2}} \right).$$

Hence (1), read literally at this finite p , is false, and false at a point where the minimum in (1) is not the cap $\log p$.

The last clause is the substance of the example: in the saturated branch (1) degenerates into the demand that $X + X'$ be exactly uniform, whereas Theorem 1 places the failure strictly *below* the saturation threshold, in the branch that carries the content $H(X) + \frac{1}{2} - o(1)$.

2. THE WITNESS, EXHIBITED

Fix $p = 101$, $m = 71$, $S = \{0, \dots, m-1\}$ and X uniform on S , so that $H(X) = \log 71 = 6.149747119504682\dots$ and

$$(3) \quad 2^{H(X)+1} - 1 = 2m - 1 = 141 \quad \text{exactly.}$$

Because X is uniform on a set, the right-hand side of (1) is an *integer* expression, and this is what makes the whole test finite and exact.

Which branch is active. By (3) the two branches compare as $(2m-1)/\sqrt{2}$ against p , i.e. as the integers $(2m-1)^2$ against $2p^2$:

$$(4) \quad (2m-1)^2 = 141^2 = 19881 < 20402 = 2 \cdot 101^2 = 2p^2.$$

So $141/\sqrt{2} < 101$ and the minimum in (1) is the second branch:

$$\begin{aligned} \log\left(\frac{141}{\sqrt{2}}\right) &= \log 141 - \frac{1}{2} = 6.639551352398794\dots \\ &< 6.658211482751795\dots = \log 101. \end{aligned}$$

Equivalently: the least m for which the arc of length m in $\mathbb{F}_{101}$ saturates, i.e. the least m with $(2m-1)^2 \geq 2p^2$, is $m = 72$, and our witness has $m = 71$, one step below that floor.

The law of $X + X'$. Put $C_k = \#\{(i, j) \in S \times S : i + j \equiv k \pmod{p}\}$, so that $\Pr[X + X' = k] = C_k/m^2$ and $\sum_k C_k = m^2$. For the arc $S = \{0, \dots, m-1\}$ with $p < 2m-1 < 2p$ one has the closed form

$$(5) \quad C_k = f(k) + f(k+p), \quad f(s) = \begin{cases} s+1, & 0 \leq s \leq m-1, \\ 2m-1-s, & m-1 < s \leq 2m-2, \\ 0, & \text{otherwise,} \end{cases}$$

since the number of ordered representations of s as a sum of two elements of $\{0, \dots, m-1\}$ over $\mathbb{Z}$ is $f(s)$, and each residue k is hit by exactly the integers k and $k+p$ in $[0, 2m-2]$. At $p = 101$, $m = 71$ this gives the multiset

$$(6) \quad \{C_k\}_{k=0}^{100} = \{41^{(42)}, 42^{(2)}, 43^{(2)}, \dots, 70^{(2)}, 71^{(1)}\},$$

that is: the value 41 occurs 42 times, each of 42, 43, $\dots$, 70 occurs twice, and 71 occurs once. There are $42 + 29 \cdot 2 + 1 = 101$ entries and they sum to $41 \cdot 42 + 2 \cdot \sum_{v=42}^{70} v + 71 = 1722 + 3248 + 71 = 5041 = 71^2$, as they must. Concretely, reading $k = 0, 1, \dots, 100$ in order, $C_k = 41$ for $0 \leq k \leq 40$, $C_k = k+1$ for $41 \leq k \leq 70$, and $C_k = 141-k$ for $71 \leq k \leq 100$. In particular $101 C_k \neq m^2$ for $k = 41$, so $X + X'$ is not uniform on $\mathbb{F}_{101}$.

For orientation only — no step of the proof uses it — $H(X + X') = 6.6305238776\dots$, so the deficit in (1) is $-9.0274748 \times 10^{-3}$ bits.

3. THE DECISIVE INEQUALITY, IN INTEGERS

Lemma 2. *Let p be prime, let $S \subseteq \mathbb{F}_p$ with $|S| = m \geq 1$, let X be uniform on S , let X' be an independent copy and let C_k be as above. Then*

$$(7) \quad H(X + X') < \log\left(\frac{2m-1}{\sqrt{2}}\right) \iff (2m^4)^{m^2} < (2m-1)^{2m^2} \left(\prod_k C_k^{C_k}\right)^2,$$

and both sides of the right-hand inequality are positive integers.

Proof. With $c_k = C_k/m^2$ and $\sum_k C_k = m^2$,

$$2^{H(X+X')} = \prod_k c_k^{-c_k} = \frac{(m^2)^{\sum_k C_k/m^2}}{(\prod_k C_k^{C_k})^{1/m^2}} = \frac{m^2}{P^{1/m^2}}, \quad P := \prod_k C_k^{C_k},$$

the products being over the k with $C_k > 0$. Hence $H(X + X') < \log((2m-1)/\sqrt{2})$ says $m^2/P^{1/m^2} < (2m-1)/\sqrt{2}$, i.e. $\sqrt{2} m^2 < (2m-1)P^{1/m^2}$. Squaring (both sides are positive) gives $2m^4 < (2m-1)^2 P^{2/m^2}$, and raising to the power m^2 gives (7). Every C_k is a nonnegative integer, so $P \in \mathbb{Z}_{>0}$. $\square$

Remark. Criterion (7) tests (1) *only after* the branch test (4) has returned “out of band”. In the saturated branch the right-hand side of (1) is $\log p$, not $\log((2m-1)/\sqrt{2})$, and there the right-hand inequality of (7) can hold with no violation whatsoever: for $S = \mathbb{F}_p$ itself, $X + X'$ is uniform, (1) holds with equality, and yet $(2m^4)^{m^2} < (2m-1)^{2m^2} P^2$ is true because it is being compared against the *inactive* branch. So $(2m-1)^2 < 2p^2$ is a guard and not a formality. It holds for the witness of Theorem 1 by (4).

Proposition 3. *At $p = 101$, $m = 71$, with C_k as in (6) and $P = \prod_k C_k^{C_k}$, the two integers $L = (2m^4)^{m^2} = (2 \cdot 71^4)^{5041}$ and $R = (2m-1)^{2m^2} P^2 = 141^{10082} P^2$ satisfy*

$$2^{91} L < R < 2^{92} L,$$

and in particular $L < R$.

Proof of Theorem 1. The branch identification is (3) and (4). The inequality $L < R$ of Proposition 3 together with Lemma 2 gives $H(X + X') < \log(141/\sqrt{2})$, which is exactly the failure of (1) at this X . $\square$

The bracket $2^{91} L < R < 2^{92} L$ is worth stating because it converts the deficit into a rational bound with no logarithm evaluated. Indeed the proof of Lemma 2 shows $\log(R/L) = 2m^2(\log((2m-1)/\sqrt{2}) - H(X + X'))$, so with $2m^2 = 10082$,

$$\frac{91}{10082} < \log\left(\frac{141}{\sqrt{2}}\right) - H(X + X') < \frac{92}{10082},$$

that is, $0.0090259\dots < \text{deficit} < 0.0091251\dots$ bits. The violation is therefore bounded away from zero by an inequality between integers.

4. FURTHER WITNESSES

The same two integer tests certify a violation for the smallest out-of-band arc at each of six more primes. In each row $S = \{0, \dots, m-1\}$, the third and fourth columns are the branch test $(2m-1)^2 < 2p^2$ of (4), and the fifth is and $k := \lfloor \log_2(R/L) \rfloor$, so that $2^k L < R < 2^{k+1} L$ as in Proposition 3. Since $k \geq 0$ in every row, $L < R$ and (1) fails at each; and by the identity displayed above the deficit at each witness lies strictly between $k/2m^2$ and $(k+1)/2m^2$ bits. The last column records that bracket to three significant figures; it is the only inexact entry in the table, and nothing depends on it.

p	m	$(2m-1)^2$	$2p^2$	k	deficit (bits)
61	43	7225	7442	21	$5.68\text{--}5.95 \cdot 10^{-3}$
101	71	19881	20402	91	$9.03\text{--}9.13 \cdot 10^{-3}$
151	106	44521	45602	241	$1.072\text{--}1.077 \cdot 10^{-2}$
211	147	85849	89042	160	$3.70\text{--}3.73 \cdot 10^{-3}$
307	213	180625	188498	37	$4.08\text{--}4.19 \cdot 10^{-4}$
401	278	308025	321602	46	$2.98\text{--}3.04 \cdot 10^{-4}$
509	353	497025	518162	316	$1.268\text{--}1.272 \cdot 10^{-3}$

The deficit column is not monotone in p , and no trend should be read into these seven rows; Section 5 returns to this. We do not claim that a violating arc exists at every prime.

5. WHAT IS NOT SETTLED

No theorem is contradicted. Every witness above has $0.50 < \log p - H(X) < 0.53$ bits, so all seven sit at the near-uniform end of the entropy range. We have not determined the constant K_ε of the window of (2) and claim nothing about its size, so we do not assert that the window excludes our witnesses; what is checked directly is that $H(X + X') - H(X) > 0.477$ at all seven witnesses (weakest certified bound 0.47701, at $p = 61$), so every one of them satisfies (2) for every $\varepsilon > 0.023$.

Only the printed finite- p form is refuted. What fails is (1) exactly as displayed, with no ε and no window; the second line of the quoted display carries an $o_{p \rightarrow \infty}(1)$ and is untouched, and a reader who takes the display to be intended asymptotically should read this note as bearing on its finite- p form alone. We make no claim about the $p \rightarrow \infty$ behaviour, and the seven rows of Section 4 support none: their deficits are not monotone in p , and the deficit of the smallest out-of-band arc may well tend to 0, in which case the second line survives intact. Nor are the rows maximal deficits, each being the *smallest* out-of-band arc at its prime. Whether the supremum of the deficit over all X stays bounded away from 0 as $p \rightarrow \infty$ is open, and we do not conjecture an answer.

The bulk regime is untouched, and no minimality is claimed. All of our witnesses sit just below the saturation floor; the regime where $2^{H(X)}$ is well below $p/\sqrt{2}$ is not addressed here at all. We do not claim that $p = 101$ or $m = 71$ is extremal in any sense, we search no prime beyond $p = 509$, and

only arcs $\{0, \dots, m-1\}$ with uniform X are considered; no other family of sets and no non-uniform X is examined.

6. REPRODUCTION

Everything needed is printed above: the prime $p = 101$, the set $S = \{0, \dots, 70\}$, the closed form (5) for the convolution counts, the resulting multiset (6), the branch test (4) and the integer criterion (7). A referee who prefers to recompute rather than to read can do so in a few lines of exact integer arithmetic. A program that does exactly that, reading only the data printed here and using the standard library alone with no floating-point decision, accompanies this note as `verify.py`, together with the recorded output `verify.output.txt` of a run of it.

REFERENCES

- [1] L. Gavalakis, M. Goh, and I. Kontoyiannis, *Entropy lower bounds and sum-product phenomena*, arXiv:2604.20233, version 3 (28 August 2026); quotations and the equation number refer to that version.
- [2] T. Tao, *Sumset and inverse sumset theory for Shannon entropy*, *Combin. Probab. Comput.* **19** (2010), no. 4, 603–639. doi:10.1017/S0963548309990642; arXiv:0906.4387.